Layer 4: Green Infrastructure and Landscape PV (LPV) Report

This report provides a detailed technical breakdown of the parameters, configurations, calculations, and modeling limitations under **Layer 4: Green Infrastructure & Landscape PV (LPV)** within the LMN Neighbourhood Design Tool.

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1. Core Objectives

Layer 4 enables stakeholders to integrate urban greening, local agricultural systems, and energy-integrated landscape photovotaics (LPV) at the neighbourhood scale. It balances local solar energy generation against urban microclimate improvements and pedestrian shading comfort.

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2. Green Infrastructure and Urban Agriculture Configurations

The tool allows selecting and overlaying three categories of green elements:

2.1 Greening Infrastructure

- **Green Roofs**: Extensive or intensive vegetative roof coverings.
- **Vertical Greening Systems**: Green walls or facades providing vertical insulation and shading.
- **Linear Greenery**: Continuous vegetative borders along pedestrian pathways.
- **Green Spaces**: Neighbourhood parks and grassed areas.

2.2 Urban Agriculture

- **Roof Gardens**: Vegetative systems optimized for localized rooftop agriculture.
- **Food Gardens**: Community plots and agricultural areas at ground level.

2.3 Energy-Integrated GI

- **PV-Green Roofs Integrated Modules**: Co-located solar panel arrays and vegetative roof layers, leveraging transpirative cooling to enhance PV efficiency.
- **Landscape PV (LPV)**: Shading structures over walking lanes, parking spaces, and bus stops.

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3. Landscape PV (LPV) Sizing and Generation Calculations

LPV sizing is based on standardized block allocations. The following parameters and values are uniform across all Neighbourhood Units (NUs) in `NUs_LPV.csv`:

3.1 Core Sizing Parameters

- **Base Block Land Area**: 21,717 m^2 (~5 acres) for all neighbourhood configurations.
- **Land Allocation**: 20% of block land area is allocated to LPV structures (equivalent to 4,343.4 m^2 or 4,046 m^2 in the simplified template).
- **Usable Area (Aperture)**: 10% of total block area is covered by active solar cells (equivalent to 2,171.7 m^2 or 2,023 m^2 in the simplified template).
- **Module Capacity**: 400W panels.
- **Installed Capacity**: 475 kWp.
- **Annual Energy Generation**: 608 MWh/year.
- **Land Use Efficiency**: 185 kWh/m^2/year (Total annual generation divided by total land area covered).

3.2 Visual Clearances & Height Parameters

Clearance levels can be adjusted based on the functional location:

- **Pedestrian**: 2.5 meters.

- **Vehicle**: 3.0 meters.
- **Service**: 4.5+ meters.

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4. Urban Heat Island (UHI) Mitigation & Pedestrian Comfort

- **Surface Temperature Reduction**: LPV structures act as a shade canopy, reducing ground-level surface temperatures by up to -14°C (25°F) under peak solar conditions.
- **Transparency Trade-offs**: Panel transparency can be configured to 25%, 50%, or 75%.
  - **Higher transparency**: Enhances visual comfort, allows more natural ambient light to reach pedestrian paths, and supports vegetation growth underneath.
  - **Lower transparency**: Restricts light transmission but maximizes active solar cell area, thereby achieving the standard 608 MWh/year energy generation target.

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5. Known Limitations

- **Static Generation Calculations**: LPV energy output is modeled as a static 608 MWh/year for all NUs, neglecting differences in local microclimatic conditions, solar shading from surrounding buildings, and spatial structural configurations.
- **Uniform Temperature Reduction**: The -14°C temperature drop is a static benchmark value rather than a dynamically simulated microclimatic metric accounting for wind patterns or local albedo.
- **Qualitative Non-Energy GI**: The benefits of vegetative infrastructure (green roofs, vertical greening) are described qualitatively. They lack active quantitative calculations for local cooling, carbon capture, or stormwater runoff mitigation.
- **Fixed Block Geometries**: Sizing relies on a fixed block footprint (21,717 m^2), which does not capture varying real-world block configurations and boundary dimensions.